Flu-Prediction

Report and analysis: updating the thesis code and comparing classical models with deep learning models

20 October 2026

Summary

- The code now runs on Python 3.12 with current versions of scikit-learn, NumPy, Biopython and PyTorch. Naive Bayes, which crashes on recent scikit-learn, was fixed.
- The original code had a bug that made every target sequence identical. This inflated the original results: with the bug, tree models reach up to 100%; with the bug fixed, they score far lower. The charts in the thesis were very likely produced with this bug.
- Three deep learning models were added (a CNN, a Transformer and a BiLSTM). All models, old and new, are trained and tested on the same splits and scored with the same metrics.
- **The deep learning models have the best overall accuracy**: 98.76–99.25% of nucleotides correct on average, against 82.22–91.67% for the classical models. In cross-validation, the Decision Tree is ahead on the two H1N1 datasets.
- **But they barely beat a model that copies last season's sequence** (98.61%). Most of their accuracy comes from copying. On the positions that mutated, the classical models do as well or better (Decision Tree 42.4% against CNN 38.2%), but at the cost of breaking many positions that did not change.
- The main limits are the size of the data (16–35 training pairs per dataset) and the way sequences are paired (by position in the file, not by ancestry). Both could be addressed with the larger data collected by the original research; see Section 6.

1. Code updates

The original code was written for Python 3.8 and the library versions of that time. The table below lists what was changed to make it run correctly today.

Problem	Effect	Fix
Naive Bayes is given decimal class labels	Recent scikit-learn stops with "Unknown label type: continuous"	The 15-digit numbers are converted to integers. They fit exactly, so nothing is lost.
`data_test[i]` instead of `data_test[j]`	Every target sequence is a copy of the last one in the file	Fixed. See Section 1.2.
Naive Bayes split by 15-digit chunk	Pieces of the same sequence end up in both training and test, and its scores are not comparable to the other models	Split by sequence pair, like the other models
Sequences of different lengths	Models fail when encoded sequences differ in size	Shorter sequences are padded with zeros
Dataset chosen by commenting lines in and out	Easy to run the wrong experiment	`--dataset` option
H1N1 sequences start at different points	Position *i* is not the same place in the gene across a pair	`--align` option. See Section 3.

1.1 Environment

Component	Version
Python	3.12.10
scikit-learn	1.9.1
NumPy	2.5.4
Biopython	1.88
PyTorch	2.14.1 (CUDA 12.6)
GPU	NVIDIA GeForce MX570 A, 4 GB

1.2 Effect of the bug on the original results

In the original `main.py`, the loop that reads the target sequences used the index of the previous loop, which was stuck at its last value. Every target was therefore the same sequence. The task became "always output this one sequence", which a model can do without using its input at all. The table shows the original code run with and without the bug, with the thesis's score ($R^2 \times 100$, labelled "accuracy" in the thesis).

Dataset	Model	Distinct targets (bug / fixed)	Score with bug	Score fixed
H1N1-HA	Decision Tree	1 / 23	100.00	-8.5×10^{23}
H1N1-NA	Extra Trees	1 / 9	100.00	-1.7×10^{25}
H3N2-HA	Extra Trees	1 / 37	92.67	-9.4×10^{28}
H3N2-NA	Random Forest	1 / 35	91.41	-4.2×10^{26}

The thesis charts show scores between 35% and 100%. When every target is the same sequence, this score can only fall between 0 and 100. With correct targets, these models give large negative values. The exact thesis numbers cannot be reproduced (the original code had no fixed random seed and used older library versions), but the range matches the version with the bug.

2. Method

2.1 Task and data

Each dataset pairs sequence *i* of an older season with sequence *i* of a newer season. A model receives the older sequence and must output the newer one. The four datasets are the ones used in the thesis:

Dataset	Input (older)	Target (newer)	Pairs	Test pairs	Mutated positions in test
H1N1-HA	2021	2019–2020	24	8	247
H1N1-NA	2019–2020	2021	24	8	204
H3N2-HA	2021	2022	50	15	188
H3N2-NA	2021	2022	50	15	96

Mutated positions are counted with `--align`. Note that the H1N1-HA dataset runs backwards in time: as in the thesis, it uses 2021 as input and 2019–2020 as target.

2.2 Models

Model	Type	Description
Copy previous season	Baseline	Predicts that the new sequence is identical to the old one. It learns nothing, so any useful model must beat it.
Decision Tree	Classical	scikit-learn regressor on the thesis encoding (default settings).

Model	Type	Description
Random Forest	Classical	23 trees, as in the thesis.
Extra Trees	Classical	3 trees, as in the thesis.
Naive Bayes	Classical	Gaussian Naive Bayes; each 15-digit chunk is one sample, as in the thesis.
CNN	Deep	Four residual blocks of 1D convolutions (kernel 9, dilations 1, 2, 4, 8; 128 channels) over single nucleotides. Each position sees about 240 nucleotides on either side.
Transformer	Deep	Four-layer Transformer encoder (4 attention heads, dimension 128) over codons (3-letter words), with learned position embeddings.
BiLSTM	Deep	Two-layer bidirectional LSTM (128 units per direction) over codons.

The classical models use the thesis encoding: A=1, T=2, G=3, C=4, with every 15 digits joined into one number. The deep models read one token per nucleotide (or per codon) and predict one nucleotide per position. All three end in the same output layer with a *copy mechanism*, an idea from NLP (CopyNet): a learned weight that favours keeping the input nucleotide. Because about 99% of a gene does not change between seasons, this lets the networks focus on the positions that do.

Training settings for all deep models: AdamW optimiser, weight decay 0.01, cosine learning-rate schedule, 200 epochs, batch size 8, learning rate 0.001 (0.0005 for the Transformer), random seed 0. No tuning was done on the test data.

2.3 Evaluation

All models use the same splits as the thesis: 2-fold cross-validation over the sequence pairs, and a hold-out split with 70% of pairs for training and 30% for testing (`random_state=50`). Most tables in this report use the hold-out test set; Section 3.5 checks the results with cross-validation. Every model is scored with:

- **Nucleotide accuracy**: the share of nucleotides predicted correctly.
- **Mutation accuracy**: accuracy only on positions where the new sequence differs from the old one. The copy baseline always scores 0% here, so this measures whether a model predicts change.
- **R^2, MAE and MSE** on the 15-digit encoding (the thesis metrics). Predictions from the deep models are converted to the same encoding before scoring.

3. Results

All results in this section use `--align`, which trims every sequence to start at its start codon (ATG). In the H1N1 files, some sequences include a few bases before the start codon and others do not, so without alignment position i is not the same place in the gene across a pair. Results without alignment are in Section 4. The best value in each column is in bold.

3.1 Nucleotide accuracy

Model	Type	H1N1-HA	H1N1-NA	H3N2-HA	H3N2-NA	Average
Copy previous season	Baseline	97.44	98.18	99.26	99.55	98.61
Decision Tree	Classical	94.54	**98.82**	83.40	89.92	91.67
Random Forest	Classical	86.92	77.30	81.29	88.54	83.51
Extra Trees	Classical	92.87	94.22	83.40	89.92	90.10
Naive Bayes	Classical	74.27	94.04	63.84	96.73	82.22
CNN	Deep	99.24	96.87	**99.30**	**99.61**	98.76
Transformer	Deep	99.29	98.80	99.30	99.60	**99.25**
BiLSTM	Deep	**99.32**	98.48	**99.30**	**99.61**	99.18

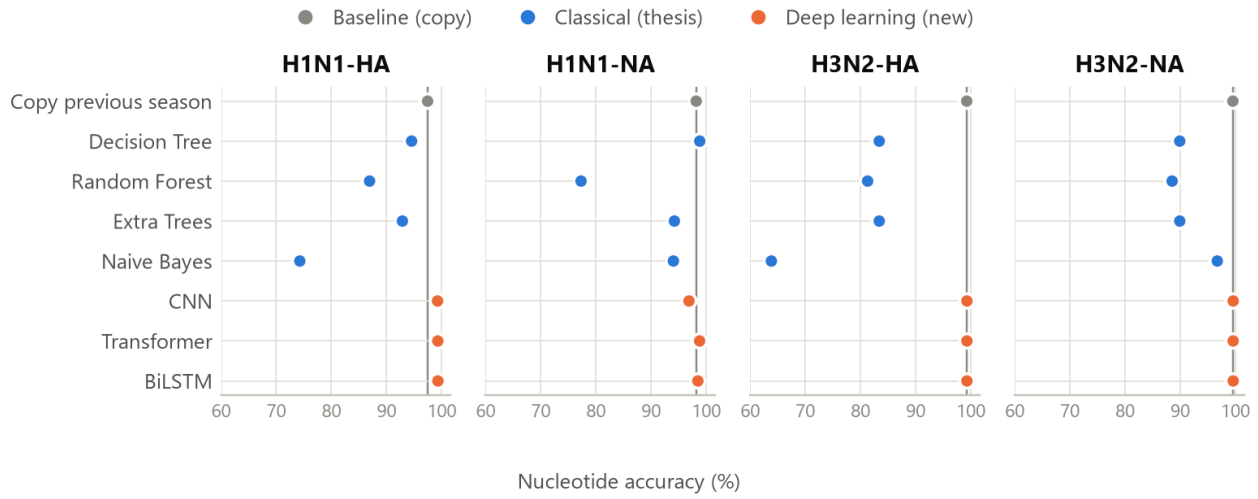


Figure 1. Nucleotide accuracy on the hold-out test set (aligned). The grey vertical line marks the copy baseline. The deep models sit on or just right of it; the classical models fall well to the left.

3.2 Mutation accuracy

Model	Type	H1N1-HA	H1N1-NA	H3N2-HA	H3N2-NA	Average
Copy previous season	Baseline	0.00	0.00	0.00	0.00	0.00
Decision Tree	Classical	66.40	63.24	24.47	**15.62**	**42.43**
Random Forest	Classical	55.06	44.61	18.09	12.50	32.56
Extra Trees	Classical	60.73	49.51	24.47	**15.62**	37.58
Naive Bayes	Classical	54.25	**63.73**	**28.72**	12.50	39.80
CNN	Deep	**68.83**	62.25	7.98	13.54	38.15
Transformer	Deep	63.97	52.45	7.45	11.46	33.83
BiLSTM	Deep	64.78	52.45	7.98	12.50	34.43

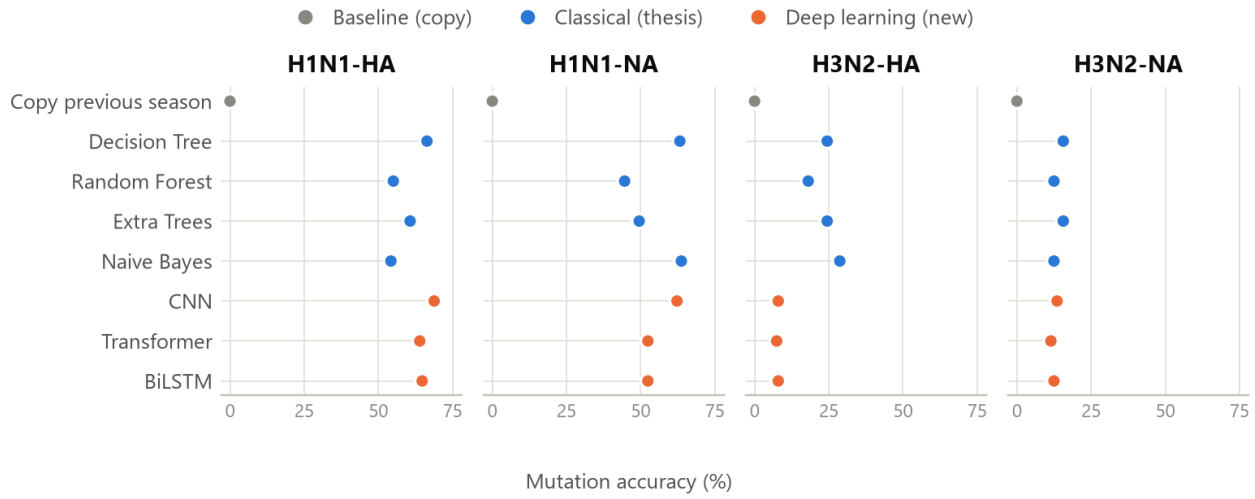


Figure 2. Mutation accuracy on the hold-out test set (aligned): the share of changed positions predicted correctly. The copy baseline is always 0%.

3.3 The trade-off between the two

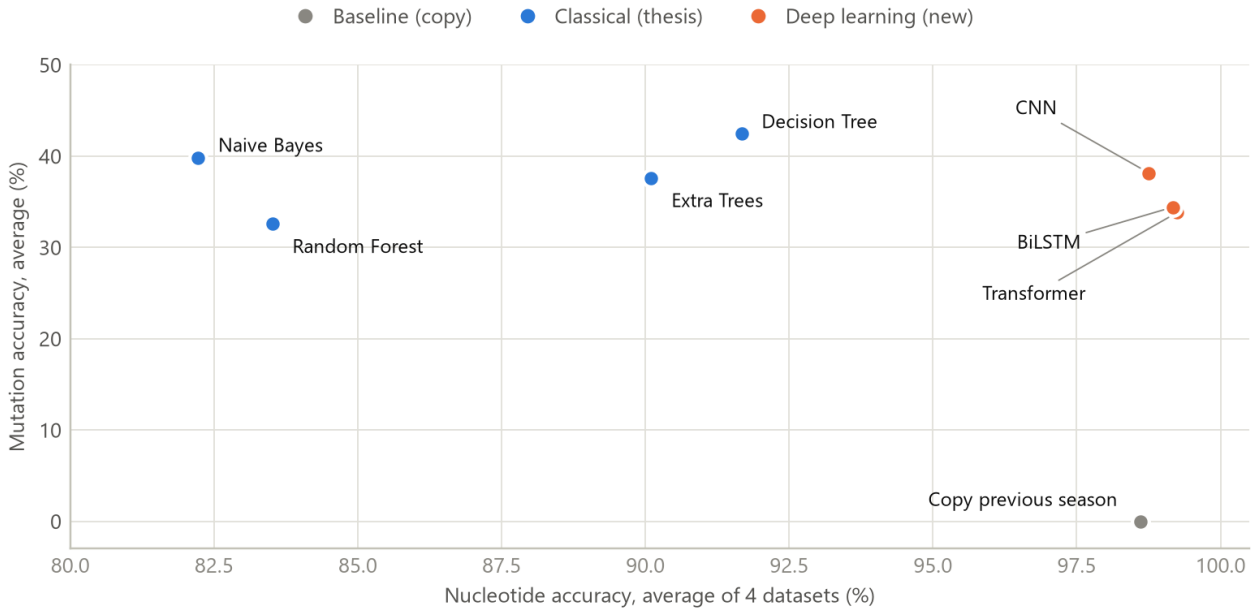


Figure 3. Average nucleotide accuracy against average mutation accuracy (aligned). Up and to the right is better. The deep models are the most accurate overall; the classical models predict a similar share of mutations but lose accuracy everywhere else.

3.4 Wrong nucleotides on the test set

Accuracy differences near 99% are easier to judge as counts. This table shows how many nucleotides each model got wrong on each hold-out test set (lower is better).

Model	Type	H1N1-HA (13,608 nt)	H1N1-NA (11,216 nt)	H3N2-HA (25,800 nt)	H3N2-NA (21,511 nt)
Copy previous season	Baseline	348	204	190	97
Decision Tree	Classical	743	**132**	4,283	2,168
Random Forest	Classical	1,780	2,546	4,826	2,466
Extra Trees	Classical	970	648	4,283	2,168
Naive Bayes	Classical	3,501	668	9,328	704
CNN	Deep	103	351	**180**	**84**
Transformer	Deep	96	135	181	85
BiLSTM	Deep	**93**	170	**180**	**84**

On H3N2-HA, for example, the best deep model makes 180 mistakes, against 190 for the copy baseline and 4,283 for the best classical model.

3.5 Cross-validation check

Nucleotide accuracy averaged over the two cross-validation folds. Each fold trains on half of the pairs, so the models have even less data than in the hold-out split.

Model	Type	H1N1-HA	H1N1-NA	H3N2-HA	H3N2-NA	Average
Copy previous season	Baseline	95.23	97.44	99.35	**99.62**	97.91
Decision Tree	Classical	**97.61**	**98.02**	86.67	85.28	91.89
Random Forest	Classical	85.02	86.06	82.02	83.81	84.23
Extra Trees	Classical	93.63	92.63	85.32	85.28	89.21
Naive Bayes	Classical	74.57	93.23	62.42	97.22	81.86
CNN	Deep	96.79	97.15	**99.36**	99.58	98.22
Transformer	Deep	96.97	97.73	99.35	99.54	**98.40**
BiLSTM	Deep	96.86	97.69	99.35	99.59	98.37

On average the deep models are still the most accurate (98.40% for the best), but the ranking changes on H1N1: with only 12 training pairs per fold, the Decision Tree is ahead of all deep models on both H1N1 datasets. The advantage of the deep models is clear only on H3N2, where it is also the smallest.

3.6 The thesis metric (R^2)

R^2 on the 15-digit encoding does not separate the models. The standard (uniform) R^2 ranges from about 0.5 to -10^{25}, because some columns of 15-digit numbers barely vary across sequences, and any error on such a column produces a huge negative value. The variance-weighted R^2 is more stable, and is at or below zero for every model:

Model	Type	H1N1-HA	H1N1-NA	H3N2-HA	H3N2-NA	Average
Copy previous season	Baseline	−1.025	**−1.133**	−1.020	−0.092	−0.817
Decision Tree	Classical	**0.000**	−1.666	−0.106	−0.070	−0.460
Random Forest	Classical	−0.121	−1.166	**−0.097**	**−0.069**	**−0.363**
Extra Trees	Classical	−0.224	−2.363	−0.106	−0.070	−0.691
Naive Bayes	Classical	−7.156	−2.718	−5.364	−0.098	−3.834
CNN	Deep	0.000	−2.115	−0.645	−0.092	−0.713
Transformer	Deep	−1.000	−1.666	−0.832	−0.092	−0.898
BiLSTM	Deep	0.000	−1.666	−0.645	−0.092	−0.601

The cause is the encoding itself: one wrong nucleotide at the start of a 15-digit chunk changes the number by trillions, while one at the end changes it by one. The error depends on where the mistake is, not on how many mistakes there are. For this reason, nucleotide and mutation accuracy are better measures for this task.

3.7 Training time

Time in seconds to train and predict once on the hold-out split. The classical models run on the CPU; the deep models on the GPU. Bold marks the slowest model in each column.

Model	Type	H1N1-HA	H1N1-NA	H3N2-HA	H3N2-NA	Average
Decision Tree	Classical	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1
Random Forest	Classical	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1
Extra Trees	Classical	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1
Naive Bayes	Classical	< 0.1	< 0.1	< 0.1	< 0.1	< 0.1
CNN	Deep	30.6	25.8	68.1	57.4	45.5
Transformer	Deep	**35.6**	**28.1**	**77.8**	**63.0**	**51.1**
BiLSTM	Deep	13.9	11.5	33.4	28.2	21.7

4. Results without alignment (thesis setup)

These results use the files exactly as in the thesis, without `--align`.

4.1 Nucleotide accuracy

Model	Type	H1N1-HA	H1N1-NA	H3N2-HA	H3N2-NA	Average
Copy previous season	Baseline	27.36	24.48	99.27	99.55	62.66
Decision Tree	Classical	94.65	**99.01**	84.89	85.40	90.99
Random Forest	Classical	85.67	75.00	82.23	84.52	81.85
Extra Trees	Classical	93.05	89.13	84.89	85.40	88.12
Naive Bayes	Classical	39.21	92.16	65.60	97.90	73.72
CNN	Deep	98.38	89.00	**99.31**	**99.62**	96.58
Transformer	Deep	99.30	94.15	99.31	99.61	**98.09**
BiLSTM	Deep	**99.32**	91.74	**99.31**	99.61	97.50

4.2 Mutation accuracy (H3N2 only)

Without alignment, thousands of H1N1 positions appear mutated only because the sequences are shifted, so mutation accuracy is meaningful only for H3N2, whose sequences were already aligned.

Model	Type	H3N2-HA	H3N2-NA
Copy previous season	Baseline	0.00	0.00
Decision Tree	Classical	20.21	16.67
Random Forest	Classical	17.55	11.46
Extra Trees	Classical	20.21	16.67
Naive Bayes	Classical	**23.40**	**21.88**
CNN	Deep	7.98	14.58
Transformer	Deep	7.45	11.46
BiLSTM	Deep	7.98	12.50

The conclusions are the same as with alignment. The Transformer has the best average nucleotide accuracy (98.09%), and the classical models generally predict more of the mutated positions. The one difference is H1N1-NA, where the Decision Tree clearly wins (99.01% against 94.15% for the Transformer). The classical models do not compare positions one by one, so the misalignment hurts them less.

5. Analysis

5.1 Do the newer models beat the older ones?

Yes, for overall accuracy. The deep models have the highest nucleotide accuracy on every dataset except H1N1-NA, where the Decision Tree (98.82%) and the Transformer (98.80%) are tied. On average, the best deep model (Transformer, 99.25%) is 7.6 points ahead of the best classical model (Decision Tree, 91.67%). In cross-validation (Section 3.5), however, the Decision Tree is ahead on both H1N1 datasets, so this advantage is solid only on H3N2 and on the average.

No, for predicting mutations. Averaged over the four datasets, the Decision Tree predicts 42.4% of mutated positions, against 38.2% for the CNN, the best deep model. On H3N2-HA, every classical model predicts more mutations than every deep model (18–29% against 7–8%).

And not by much against copying. The best deep model is only 0.64 points above the copy baseline (98.61%). In practice, the deep models learn to copy the input and change a small number of positions.

5.2 Why the classical models predict more mutations

The classical models do not learn to transform the input sequence. They learn to output something close to a typical sequence of the newer season. That output contains many of the newer season's mutations, so it scores well on mutated positions. But it also overwrites positions where the input was already correct, which is why their overall accuracy drops to 82–92%. The deep models make the opposite trade: they keep almost everything and change little.

5.3 Why the deep models cannot do better here

- **Very little data.** Each model trains on only 16–35 sequence pairs. Deep learning needs far more examples to learn patterns of evolution rather than memorise the training set.
- **The pairs are arbitrary.** Sequence *i* of the older season is not the ancestor of sequence *i* of the newer season; they are matched only by their order in the file. Many "mutations" in a pair are differences between unrelated strains, which no model can predict from the input.
- **H1N1-HA runs backwards in time**, so it measures prediction of the past rather than the future.

5.4 Limitations of this evaluation

- The test sets are small (8 pairs for H1N1, 15 for H3N2), so differences of a few mutated positions can change the ranking.
- Each deep model was trained once, with one random seed, and without hyperparameter tuning.
- The thesis models keep their original settings (for example, 3 trees for Extra Trees) to stay comparable with the thesis.

6. Recommended next steps

- **Pair sequences by similarity.** Pair each newer sequence with its most similar older sequence (or use a phylogenetic tree), so each pair reflects real evolution.
- **Use the larger dataset collected by the original research**, which holds far more sequences, for example 1,249 H3N2-HA sequences from 2021 and 2,633 from 2022. With thousands of pairs, the deep models would have a fair chance to show an advantage.
- **Report nucleotide and mutation accuracy** instead of R^2 on the 15-digit encoding.
- **Repeat each run with several random seeds** and report the mean and spread.
- **Consider a pretrained DNA language model** (for example, Nucleotide Transformer or DNABERT-2), which brings knowledge from millions of sequences and suits small datasets.

Appendix: reproducing the results

From the project folder, with the virtual environment created as described in README.md:

```
python main.py --dataset all --align
python main.py --dataset all
```

Each run takes about 20 minutes on the GPU listed in Section 1.1 and writes one CSV per dataset to `results/` (`*-aligned.csv` for the aligned run). All tables and charts in this report are generated from those files.